Symbiosis Institute of Technology, Pune

Faculty of Engineering
CSE- Academic Year 2026-27
Java Programming – Lab Batch 2024-28

Lab Assignment No:- 21	
Name of Student	Abishek Yadav
PRN No.	25070122014
Batch	A1
Class	CSE
Academic Year & Semester	2026-2027/3RD
Date of Submission	07-10-26
Title of Assignment:	To demonstrate and explore the JDBC classes and interfaces DriverManager, Connection, and Statement for establishing database connectivity and executing SQL statements.
Theory:	The JDBC API consists of several classes and interfaces that facilitate database connectivity. The DriverManager class is responsible for loading the appropriate JDBC driver and establishing a connection with the database using a database URL, username, and password. The Connection interface represents an active session between the Java application and the database server. The Statement interface is used to execute SQL queries on the connected database.
Practice Questions	1. Create a program to connect Java with database and display database connection status.

CODE:

```java
import java.sql.Connection;
import java.sql.DriverManager;

public class DatabaseConnection
{
    public static void main(String[] args)
    {
        String url = "jdbc:mysql://localhost:3306/school";
        String username = "root";
        String password = "abishek@123";

        try
        {
            Connection con = DriverManager.getConnection(
                url, username, password);
            System.out.println(
                "Database connected successfully!");
            con.close();
        }
        catch (Exception e)
        {
            System.out.println("Database connection failed!");
            System.out.println(e.getMessage());
        }
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac -cp "lib\mysql-connector-j-26.7.0.jar" DatabaseConnection.java
C:\Users\Bina Computer\OneDrive\Documents\Java>java -cp ".;lib\mysql-connector-j-26.7.0.jar" DatabaseConnection
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Database connected successfully!
```

QUESTION	2. Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.

CODE:

```java
import java.sql.Connection;
import java.sql.DriverManager;

public class StudentDatabaseConnection
{
    public static void main(String[] args)
    {
        String url = "jdbc:mysql://localhost:3306/school";
        String username = "root";
        String password = "abishek@123";

        try
        {
            Connection con = DriverManager.getConnection(
                url, username, password);
            System.out.println(
                "Database connected successfully!");
            System.out.println(
                "Student database is connected successfully.");
            con.close();
        }
        catch (Exception e)
        {
            System.out.println("Database connection failed!");
            System.out.println(e.getMessage());
        }
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac -cp "lib\mysql-connector-j-
26.7.0.jar" StudentDatabaseConnection.java
C:\Users\Bina Computer\OneDrive\Documents\Java>java -cp ".;lib\mysql-connector-j-
26.7.0.jar" StudentDatabaseConnection
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Database connected successfully!
Student database is connected successfully.
```

Conclusion	The Java programs successfully connected to the MySQL database using JDBC. The DriverManager class and Connection interface were used to establish the connection, and the database connection status was displayed successfully. Exception handling using try-catch was used to report connection failures.